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grupa 131

Examen CDI

Subiectul 2

$f : \mathbb{R}^2 \to \mathbb{R}$

$f(x,y) = x^4 - 4xy^3 + 6y^2 \quad (\forall)(x,y) \in \mathbb{R}^2$

f este obț prin operații cu fct elem $\Rightarrow$ f fct cont pe $\mathbb{R}^2 \Rightarrow D_f = \phi$

$\frac{\partial f}{\partial x}(x,y) = (x^4 - 4xy^3 + 6y^2)'_x = 4x^3 - 4y^3 \quad (\forall)(x,y) \in \mathbb{R}^2$

$\frac{\partial f}{\partial y}(x,y) = (x^4 - 4xy^3 + 6y^2)'_y = -12xy^2 + 12y \quad (\forall)(x,y) \in \mathbb{R}^2$

$\left.\begin{array}{l} \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \text{ fct cont pe } \mathbb{R}^2 \\ \mathbb{R}^2 \text{ mulț deschisă} \end{array}\right\} \Rightarrow \begin{array}{l} f \in C^1(\mathbb{R}^2) \\ D_1 = \phi \end{array}$

$(\mathbb{R}^2) \begin{cases} \frac{\partial f}{\partial x}(x,y) = 0 \\ \frac{\partial f}{\partial y}(x,y) = 0 \end{cases}$

$\begin{cases} 4x^3 - 4y^3 = 0 \\ -12xy^2 + 12y = 0 \end{cases} \Rightarrow \begin{cases} x^3 = y^3 \\ xy^2 - y = 0 \end{cases} \Rightarrow$

$\begin{cases} x = y \\ y(xy - 1) = 0 \end{cases} \Rightarrow \begin{array}{l} y = 0 \Rightarrow x = 0 \\ xy - 1 = 0 \Rightarrow xy = 1 \Rightarrow \end{array} \begin{array}{l} x = -1, y = -1 \\ x = 1, y = 1 \end{array}$

$(0,0), (1,1), (-1,-1) \in \mathbb{R}^2 \Rightarrow C = \{(0,0); (1,1); (-1,-1)\}$

$\frac{\partial^2 f}{\partial x^2}(x,y) = (4x^3 - 4y^3)'_x = 12x^2 \quad (\forall)(x,y) \in \mathbb{R}^2$

$\frac{\partial^2 f}{\partial x \partial y}(x,y) = (-12xy^2 + 12y)'_x = -12y^2 \quad (\forall)(x,y) \in \mathbb{R}^2$

$$\frac{\partial^2 f}{\partial y \partial x}(x,y) = (4x^3 - 4y^3)'_y = -12y^2 \quad (\forall)(x,y) \in \mathbb{R}^2$$

$$\frac{\partial^2 f}{\partial y^2}(x,y) = (-12xy^2 + 12y)'_y = -24xy + 12 \quad (\forall)(x,y) \in \mathbb{R}^2$$

$$\left.\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}, \frac{\partial^2 f}{\partial y^2} \text{ fct cont pe } \mathbb{R}^2 \right\} \Rightarrow \begin{array}{l} f \in C^2(\mathbb{R}^2) \\ D_2 = \phi \end{array}$$

$\mathbb{R}^2$ mult. desch.

$$Hf(0,0) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2}(0,0) & \frac{\partial^2 f}{\partial x \partial y}(0,0) \\ \frac{\partial^2 f}{\partial y \partial x}(0,0) & \frac{\partial^2 f}{\partial y^2}(0,0) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 12 \end{pmatrix}$$

$$\left.\begin{array}{l} \Delta_1 = 0 \\ \Delta_2 = 0 \end{array}\right\} \text{ nu ne putem pronunța}$$

$$Hf(1,1) = \begin{pmatrix} 12 & 12 \\ 12 & -12 \end{pmatrix}$$

$$\left.\begin{array}{l} \Delta_1 = 12 > 0 \\ \Delta_2 = -12^2 - 12^2 < 0 \end{array}\right\} \overset{\text{crt}}{\underset{\text{Sylvester}}{\Rightarrow}} (1,1) \text{ nu este pct de ext loc}$$

$$Hf(-1,-1) = \begin{pmatrix} 12 & 12 \\ 12 & -12 \end{pmatrix}$$

$$\left.\begin{array}{l} \Delta_1 = 12 > 0 \\ \Delta_2 = -12^2 - 12^2 < 0 \end{array}\right\} \overset{\text{crt}}{\underset{\text{Sylvester}}{\Rightarrow}} (-1,-1) \text{ nu este pct de ext loc}$$

$D_3 = \{(1,1), (-1,-1)\}$

$D_4 = \{(0,0)\}$

$f(x,y) - f(0,0) = x^4 - 4xy^3 + 6y^2 - 0 = x^4 - 4xy^3 + 6y^2$

$f\left(\frac{1}{n}, 0\right) = \left(\frac{1}{n}\right)^4 - 0 + 0 = \left(\frac{1}{n}\right)^4 > 0$

$\cancel{f\left(\frac{1}{n}, \frac{1}{n}\right) = \left(\frac{1}{n}\right)^4 - 4\frac{1}{n}}$

$f(x,y) = x^4 - 4xy^3 + 6y^2 = x^4 - xy^3 - 3xy^3 + 6y^2$

$$= x(x^3 - y^3) \underbrace{- 3\underbrace{\underbrace{y^2}_{>0}\underbrace{(xy-2)}_{<0}}_{<0}}_{>0}$$

pt două subșiruri ce tind la 0 :

$-3y^2(xy-2) > x(x^3 - y^3)$

pt orice 2 subșiruri $(x_n)_{n\in\mathbb{N}}, (y_n)_{n\in\mathbb{N}}$ cu $x_n \to 0$, $y_n \to 0$

$\left.\begin{array}{l} f(x_n, y_n) \geq 0 \\ \cancel{f(x,y) - f(0,0) \geq 0} \end{array}\right\} \Rightarrow$ minim local $(0,0)$

$E = \{(0,0)\}$

Subiectul 1.

$$\sum_{n=1}^{\infty} \left(\frac{an}{n+1}\right)^{n(n+1)}, \ a>0$$

$$x_n = \left(\frac{an}{n+1}\right)^{n(n+1)}, \ a>0$$

$x_n > 0 \ (\forall) n \in \mathbb{N}^*$

$$l = \lim_{n\to+\infty} \sqrt[n]{x_n} = \lim_{n\to\infty} \sqrt[n]{\left(\frac{an}{n+1}\right)^{n(n+1)}}$$

$$= \lim_{n\to+\infty} \left(\frac{an}{n+1}\right)^{n+1}$$

$$= \lim_{n\to+\infty} \left(1 + \frac{an}{n+1} - 1\right)^{n+1}$$

$$= \lim_{n\to+\infty} \left(1 + \frac{an-n-1}{n+1}\right)^{n+1}$$

$$= \lim_{n\to\infty} \left(1 + \frac{n(a-1)-1}{n+1}\right)^{n+1}$$

$$= \lim_{n\to\infty} \left[\left(1 + \frac{n(a-1)-1}{n+1}\right)^{\frac{n+1}{n(a-1)-1}}\right]^{\frac{n(a-1)-1}{\cancel{n+1}} \cdot \cancel{n+1}}$$

$$= \lim_{n\to\infty} \underbrace{\left[\left(1 + \frac{n(a-1)-1}{n+1}\right)^{\frac{n+1}{n(a-1)-1}}\right]}_{\to e}^{n(a-1)-1}$$

$$\lim_{n\to\infty} n(a-1)-1 = \begin{cases} -1, & a=1 \\ -\infty, & a\in(0,1) \\ +\infty, & a\in(1,+\infty) \end{cases}$$

$$\lim_{n\to\infty} \left[\left(1 + \frac{n(a-1)-1}{n+1}\right)^{\frac{n+1}{n(a-1)-1}}\right]^{n(a-1)-1} = \begin{cases} 0, & a\in(0,1) \\ \frac{1}{e}, & a=1 \\ \infty, & a\in(1,+\infty) \end{cases}$$

~~t~~ pt $a \in (0,1) \Rightarrow l = 0 < 1 \Rightarrow$ serie convergentă

pt $a = 1 \Rightarrow l = \frac{1}{e} < 1 \Rightarrow$ serie ~~de~~ convergentă

pt $a \in (1, +\infty) \Rightarrow l = +\infty > 1 \Rightarrow$ serie divergentă

$\Rightarrow$ pt $a \in (0, 1] \Rightarrow \sum_{n=1}^{+\infty} x_n$ serie convergentă

$a \in (1, +\infty) \Rightarrow \sum_{n=1}^{+\infty} x_n$ serie divergentă

Subiectul 3

$$3^x > 1 + \frac{x \ln 3}{1!} + \frac{x^2 (\ln 3)^2}{2!} + \frac{x^3 (\ln 3)^3}{3!} + \frac{x^4 (\ln 3)^4}{4!} \quad (\forall)\ x \in (0, +\infty)$$

$$3^x > \sum_{k=0}^{4} \frac{(\ln 3)^k}{k!} \cdot x^k$$

Fie $f : [0, +\infty) \to \mathbb{R}$

$f(x) = 3^x$

f fct de cls C^∞ pe $\mathbb{R}_+$ $\Rightarrow$ f deriv de 5 ori pe $\mathbb{R}_+$

$x_0 = 0 \in (-\infty, 0]$

$\mathbb{R}_+$ - interval nedegenerat

$x_0 = 0 \in \mathbb{R}_+$

apl formula lui Taylor cu restul sub forma lui Lagrange:

$(\forall)\ x \in [0, +\infty)$ cu $x \neq 0$ $(\exists)\ c \in [0, +\infty)$ sit între 0 și x a.î.:

$f(x) = T_{f,4,0} + R_{f,4,0}$

$$T_{f,4,0} = \sum_{k=0}^{4} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

$$= \sum_{k=0}^{4} \frac{3^0 \cdot (\ln 3)^k}{k!} x^k$$

$$= \sum_{k=0}^{4} \frac{(\ln 3)^k}{k!} x^k$$

$$R_{f,4,0} = \frac{3^c (\ln 3)^{n+1}}{(n+1)!} \cdot x^{n+1}$$

obs că $R_{f,4,0} \geq 0$ $(\forall)$ $x \in [0, \infty)$, iar egalitatea se reali-

1 zează doar pentru $x = 0$

$\Rightarrow R_{f,4,0} > 0$ pt $x \in (0, +\infty)$

$$f(x) = T_{f,4,0} + R_{f,4,0}$$

$$= \sum_{k=0}^{4} \frac{(\ln 3)^k}{k!} \cdot x^k + \frac{3^c (\ln 3)^{n+1}}{(n+1)!} \cdot x^{n+1} \Rightarrow$$

(2) $$\sum_{k=0}^{4} \frac{(\ln 3)^k}{k!} \cdot x^k + \frac{3^c (\ln 3)^{n+1}}{(n+1)!} x^{n+1} \geq \sum_{k=0}^{4} \frac{(\ln 3)^k}{k!} \cdot x^k \quad (\forall)\, x \in [0, +\infty)$$

(1),(2) $$\Rightarrow \sum_{k=0}^{4} \frac{(\ln 3)^k}{k!} \cdot x^k + \frac{3^c (\ln 3)^{n+1}}{(n+1)!} x^{n+1} > \sum_{k=0}^{4} \frac{(\ln 3)^k}{k!} \cdot x^k \quad (\forall)\, x \in (0, +\infty)$$

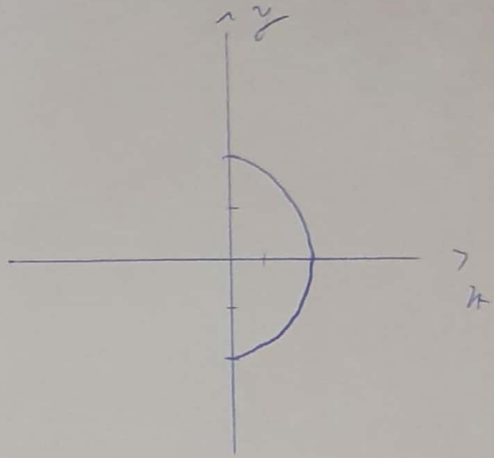
$$\Rightarrow 3^x > \sum_{k=0}^{4} \frac{(\ln 3)^k}{k!} \cdot x^k \quad (\forall)\, x \in (0, +\infty)$$

Subiectul 4

a) $\iint_D xy\,dx\,dy$

$D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 4,\ x \ge 0\}$

realizăm gf



$\begin{cases} x^2+y^2\le 4 \\ x\ge 0\end{cases} \Rightarrow \begin{cases} x^2 \le 4-y^2 \\ x \ge 0\end{cases}$

$\begin{cases} x^2\le 4\\ x\ge 0\end{cases} \Rightarrow \begin{cases} -2\le x\le 2\\ x\ge 0\end{cases} \Rightarrow x\in[0,2]$

$\cancel{0\le}\ x^2+y^2 \le 4 \Rightarrow y^2\le 4-x^2 \Rightarrow |y| \le \sqrt{4-x^2}$

$\cancel{0 \le y^2 \le 4-x^2 \Rightarrow y \in}$

$-\sqrt{4-x^2} \le y \le \sqrt{4-x^2}$

$D = \{(x,y)\in\mathbb{R}^2 \mid x\in[0,2],\ \underbrace{-\sqrt{4-x^2}}_{\alpha} \le y \le \underbrace{\sqrt{4-x^2}}_{\beta}\}$

Fie $\alpha,\beta:[0,2]\to\mathbb{R}$

$\alpha(x) = -\sqrt{4-x^2}$ $(\forall)$ $x\in[0,2]$

$\beta(x)=\sqrt{4-x^2}$ $(\forall)$ $x\in[0,2]$

α,β fct cont pe $[0,2]$

D mulț simplă în rap cu axa Oy $\Rightarrow D \in J(\mathbb{R}^2)$

Fie $f: D\to\mathbb{R}$

$f(x,y) = xy$

f fct cont pe D

D mulț simplă în rap cu axa Oy $\Big\}$ $\Rightarrow$ f fct integrabilă Riemann pe D

$$\iint_D f(x,y)\,dx\,dy = \int_0^2 \left(\int_{\varphi(x)}^{\psi(x)} f(x,y)\,dy \right) dx$$

$$= \int_0^2 \left(\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} x y\,dy \right) dx$$

$$\cancel{= \int_0^2 \left(\int_{-\sqrt{4-x^2}} \right.}$$

$$= \int_0^2 \left(\frac{x y^2}{2} \Bigg|_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \right) dx$$

$$= \int_0^2 \frac{x}{2} \left[\left(\sqrt{4-x^2}\right)^2 - \left(-\sqrt{4-x^2}\right)^2 \right] dx$$

$$= \int_0^2 \frac{x}{2} \left[(4-x^2) - (4-x^2) \right] dx$$

$$= \int_0^2 \frac{x}{2} \cdot 0 \, dx$$

$$= \int_0^2 0 \, dx = 0$$

b) $f : [0, \infty) \to [0, \infty)$

$f(0) = 0$

f cont

$|f(x) - f(y)| \geq |\sqrt{x} - \sqrt{y}|$ $(\forall)$ $x, y \in [0, \infty)$

$$\underbrace{|f(x) - f(y)| - |\sqrt{x} - \sqrt{y}|}_{g(x)} \geq 0$$

Fie $g : \mathbb{R} \to [0, \infty)$

$g(x) = |f(x) - f(y)| - |\sqrt{x} - \sqrt{y}|$